Day 16: Kubernetes Deployments & Rollouts

COVER SLIDE / OVERVIEW

```
ubuntu@master-node:~/day16-rollouts

# Rolling Updates, History, Failures & Rollbacks
$ kubectl annotate deployment/nginx-deployment kubernetes.io/change-
cause="..."
$ kubectl set image deployment/nginx-deployment nginx=nginx:1.28
$ kubectl set image deployment/nginx-deployment nginx=nginx:1.99-broke
* Status: Rollback & Bonus Strategy Verified
```

MISSION ROADMAP (01 TO 07)

- ✓ **01: Rollout Concepts & Commands**
- ✓ **02: Baseline State & Strategy Inspection**
- ✓ **03: Change Annotation & Controlled Release**
- ✓ **04: Failure Induction & Event Diagnostics**
- ✓ **05: Rollback Recovery & Verification**
- ✓ **06: Bonus Strategy (maxSurge & maxUnavailable)**
- ✓ **07: Interview Readiness Q&A**

01 / Core Concepts & Rollout Commands

Deployment Rollout Concepts

- **Rolling Update:** Gradually replaces old Pods with new ones from an updated template with zero or controlled downtime.
- **ReplicaSet Revisions:** Deployments manage ReplicaSets for each template version, retaining history for rollbacks.
- **maxSurge:** Maximum extra Pods allowed above desired replica count during updates (default 25%).
- **maxUnavailable:** Maximum desired Pods allowed to be unavailable during updates (default 25%).
- **Change-Cause Annotation:** Records descriptive notes explaining why a release was triggered.
- **Rollback:** Reverts the deployment back to a previously saved stable revision.

Essential kubectl Rollout Commands

- `kubectl rollout status deployment/nginx-deployment`
- `kubectl rollout history deployment/nginx-deployment`
- `kubectl annotate deployment/nginx-deployment kubernetes.io/change-cause="..."`
- `kubectl set image deployment/nginx-deployment nginx=nginx:1.28`
- `kubectl rollout undo deployment/nginx-deployment`
- `kubectl describe pod <pod-name>` • `kubectl get rs,pods -o wide`

02 / Baseline State & Strategy Inspection (Steps 1-2)

Steps 1–2: Applied Deployment manifest (`nginx:1.27`, 3 replicas), verified active resources (`get deployments`, `get pods`, `get rs`), and reviewed strategy settings (`maxSurge: 25%`, `maxUnavailable: 25%`).

```
ubuntu@master-node:~$ nano deployment.yaml
ubuntu@master-node:~$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
ubuntu@master-node:~$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:1.27
          ports:
            - containerPort: 80
ubuntu@master-node:~$ kubectl config current-context
kubernetes-admin@kubernetes
ubuntu@master-node:~$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   3/3     3            3           63s
ubuntu@master-node:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-7c977bd7bb-5z4qr   1/1     Running   0          72s
nginx-deployment-7c977bd7bb-p89rr   1/1     Running   0          72s
nginx-deployment-7c977bd7bb-q2ntr   1/1     Running   0          72s
ubuntu@master-node:~$ kubectl get rs
NAME                          DESIRED   CURRENT   READY   AGE
nginx-deployment-7c977bd7bb   3         3         3       79s
ubuntu@master-node:~$
```

```
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl describe deployment ng
inx-deployment
Name:                   nginx-deployment
Namespace:              default
CreationTimestamp:      Mon, 19 Oct 2026 12:27:19 +0000
Labels:                 app=nginx
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=nginx
Replicas:               3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=nginx
  Containers:
   nginx:
    Image:        nginx:1.27
    Port:         80/TCP
    Host Port:    0/TCP
    Environment:  <none>
    Mounts:       <none>
  Volumes:        <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   nginx-deployment-7c977bd7bb (3/3 replicas created)
Events:
  Type    Reason             Age                   From                   Message
  ----    ------             ----                  ----                   -------
  Normal  ScalingReplicaSet  2m13s (x2 over 4m53s)  deployment-controller  Scaled up replica set nginx-
deployment-7c977bd7bb to 3
```

03 / Change Annotation & Controlled Release (Step 3)

Step 3: Applied change-cause annotation, updated image to `nginx:1.28`, verified successful rollout status, and tracked revision history.

```
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl set image deployment/n
ginx-deployment \
 nginx=nginx:1.28
deployment.apps/nginx-deployment image updated
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl rollout status deploym
ent/nginx-deployment
deployment "nginx-deployment" successfully rolled out
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl rollout history deploy
ment/nginx-deployment
deployment.apps/nginx-deployment
REVISION  CHANGE-CAUSE
1         Day 16: update Nginx for rollout practice
2         Day 16: update Nginx for rollout practice

ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get rs
NAME                          DESIRED   CURRENT   READY   AGE
nginx-deployment-7c977bd7bb   0         0         0       29m
nginx-deployment-859f76c7fc   3         3         3       99s
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-859f76c7fc-bc972   1/1     Running   0          2m1s
nginx-deployment-859f76c7fc-kbp7t   1/1     Running   0          2m2s
nginx-deployment-859f76c7fc-ka2xz   1/1     Running   0          2m7s
```


```
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl set image deployment/n
ginx-deployment \
 nginx=nginx:1.28
deployment.apps/nginx-deployment image updated
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl rollout status deploym
ent/nginx-deployment
deployment "nginx-deployment" successfully rolled out
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl rollout history deploy
ment/nginx-deployment
deployment.apps/nginx-deployment
REVISION  CHANGE-CAUSE
1         Day 16: update Nginx for rollout practice
2         Day 16: update Nginx for rollout practice

ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get rs
NAME                          DESIRED   CURRENT   READY   AGE
nginx-deployment-7c977bd7bb   0         0         0       29m
nginx-deployment-859f76c7fc   3         3         3       99s
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-859f76c7fc-bc972   1/1     Running   0          2m1s
nginx-deployment-859f76c7fc-kbp7t   1/1     Running   0          2m2s
nginx-deployment-859f76c7fc-ka2xz   1/1     Running   0          2m7s
```

04 / Failure Induction & ImagePullBackOff (Steps 5-6)

Steps 5–6: Recorded known-good revision (`nginx:1.28`), then triggered a temporary failed rollout using invalid image tag `nginx:1.99-broke`.

```
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl rollout history deploy
ment/nginx-deployment
deployment.apps/nginx-deployment
REVISION  CHANGE-CAUSE
1         Day 16: update Nginx for rollout practice
2         Day 16: update Nginx for rollout practice

ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get deployment nginx-d
eployment \
  -o jsonpath='{.spec.template.spec.containers[0].image}'; echo
nginx:1.28
```

```
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl set image deployment/n
ginx-deployment \
  nginx=nginx:1.99-broke
deployment.apps/nginx-deployment image updated
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get pods -w
NAME                                READY   STATUS         RESTARTS   AGE
nginx-deployment-64f64fdcb5-47b5z   0/1     ErrImagePull   0          29s
nginx-deployment-859f76c7fc-bc972   1/1     Running        0          6m21s
nginx-deployment-859f76c7fc-kbp7t   1/1     Running        0          6m22s
nginx-deployment-859f76c7fc-kg2xz   1/1     Running        0          6m27s
nginx-deployment-64f64fdcb5-47b5z   0/1     ImagePullBackOff   0          38s
nginx-deployment-64f64fdcb5-47b5z   0/1     ErrImagePull       0          50s
nginx-deployment-64f64fdcb5-47b5z   0/1     ImagePullBackOff   0          65s
```

04 / Failure Diagnostics & Cluster Events (Step 7)

Step 7: Inspected failing pod status (`ImagePullBackOff`) using `kubectl describe pod` and tracked scheduling timeline events.

```
^Cubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get pods
NAME                                READY   STATUS         RESTARTS   AGE
nginx-deployment-64f64fdcb5-47b5z   0/1     ErrImagePull   0          96s
nginx-deployment-859f76c7fc-bc972   1/1     Running        0          7m28s
nginx-deployment-859f76c7fc-kbp7t   1/1     Running        0          7m29s
nginx-deployment-859f76c7fc-kg2xz   1/1     Running        0          7m34s
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl describe pod nginx-dep
loyment-64f64fdcb5-47b5z
Name:             nginx-deployment-64f64fdcb5-47b5z
Namespace:        default
Priority:         0
Service Account:  default
Node:             slave-node/172.31.27.110
Start Time:       Mon, 19 Oct 2026 13:04:11 +0000
Labels:           app=nginx
                  pod-template-hash=64f64fdcb5
Annotations:      <none>
Status:           Pending
IP:               10.244.1.33
IPs:
  IP:           10.244.1.33
Controlled By:  ReplicaSet/nginx-deployment-64f64fdcb5
Containers:
  nginx:
    Container ID:
    Image:          nginx:1.99-broke
    Image ID:
    Port:           80/TCP
    Host Port:      0/TCP
    State:          Waiting
      Reason:       ImagePullBackOff
    Ready:          False
    Restart Count:  0
    Environment:    <none>
    Mounts:
```

```
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get events --sort-by=.
metadata.creationTimestamp
LAST SEEN   TYPE      REASON              OBJECT                                  MESSAGE
47m         Normal    ScalingReplicaSet   deployment/web                          Scaled up replica se
t web-5d79b66554 to 3
57m         Normal    Scheduled           pod/nginx-deployment-7c977bd7bb-hzmrl   Successfully assigne
d default/nginx-deployment-7c977bd7bb-hzmrl to slave-node
57m         Normal    SuccessfulCreate    replicaset/nginx-deployment-7c977bd7bb  Created pod: nginx-d
eployment-7c977bd7bb-h4pgl
57m         Normal    SuccessfulCreate    replicaset/nginx-deployment-7c977bd7bb  Created pod: nginx-d
eployment-7c977bd7bb-hzmrl
57m         Normal    SuccessfulCreate    replicaset/nginx-deployment-7c977bd7bb  Created pod: nginx-d
eployment-7c977bd7bb-gqmpr
57m         Normal    Scheduled           pod/nginx-deployment-7c977bd7bb-h4pgl   Successfully assigne
d default/nginx-deployment-7c977bd7bb-h4pgl to slave-node
57m         Normal    Pulled              pod/nginx-deployment-7c977bd7bb-gqmpr   Container image "ngi
nx:1.27" already present on machine
57m         Normal    Scheduled           pod/nginx-deployment-7c977bd7bb-gqmpr   Successfully assigne
d default/nginx-deployment-7c977bd7bb-gqmpr to slave-node
57m         Normal    ScalingReplicaSet   deployment/nginx-deployment             Scaled up replica se
t nginx-deployment-7c977bd7bb to 3
57m         Normal    Created             pod/nginx-deployment-7c977bd7bb-gqmpr   Created container: n
ginx
57m         Normal    Created             pod/nginx-deployment-7c977bd7bb-hzmrl   Created container: n
ginx
57m         Normal    Started             pod/nginx-deployment-7c977bd7bb-h4pgl   Started container ng
inx
57m         Normal    Created             pod/nginx-deployment-7c977bd7bb-h4pgl   Created container: n
ginx
57m         Normal    Pulled              pod/nginx-deployment-7c977bd7bb-hzmrl   Container image "ngi
nx:1.27" already present on machine
57m         Normal    Pulled              pod/nginx-deployment-7c977bd7bb-h4pgl   Container image "ngi
nx:1.27" already present on machine
57m         Normal    Started             pod/nginx-deployment-7c977bd7bb-gqmpr   Started container ng
inx
57m         Normal    Started             pod/nginx-deployment-7c977bd7bb-hzmrl   Started container ng
```

05 / Rollback Recovery & Verification (Step 8)

Step 8: Executed `kubectl rollout undo deployment/nginx-deployment` to successfully restore the stable `nginx:1.28` image and healthy running pods.

```
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl rollout history deploy
ment/nginx-deployment
deployment.apps/nginx-deployment
REVISION  CHANGE-CAUSE
1         Day 16: update Nginx for rollout practice
2         Day 16: update Nginx for rollout practice
3         Day 16: update Nginx for rollout practice
```

```
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl rollout undo deploymen
t/nginx-deployment
deployment.apps/nginx-deployment rolled back
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl rollout status deploym
ent/nginx-deployment
deployment "nginx-deployment" successfully rolled out
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl rollout history deploy
ment/nginx-deployment
deployment.apps/nginx-deployment
REVISION  CHANGE-CAUSE
1         Day 16: update Nginx for rollout practice
3         Day 16: update Nginx for rollout practice
4         Day 16: update Nginx for rollout practice

ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get deployment nginx-d
eployment -o wide
NAME               READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS   IMAGES       SELECTOR
nginx-deployment   3/3     3            3           72m   nginx        nginx:1.28   app=nginx
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-859f76c7fc-bc972   1/1     Running   0          41m
nginx-deployment-859f76c7fc-kbp7t   1/1     Running   0          41m
nginx-deployment-859f76c7fc-kg2xz   1/1     Running   0          41m
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$
```

06 / Bonus Challenge — Custom Strategy (`maxSurge: 1`, `maxUnavailable: 0`)

SECTION 06 / BONUS MANIFEST

Bonus Challenge: Configured a custom rolling strategy in `nginx-strategy-test.yaml` ensuring zero unavailability (`maxUnavailable: 0`) while allowing one extra surge pod (`maxSurge: 1`).

```
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ ls
deployment.yaml  nginx-strategy-test.yaml
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ ^C
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ nano nginx-strategy-test.yaml
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ cat nginx-strategy-test.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-strategy-test
spec:
  replicas: 3
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 0
  selector:
    matchLabels:
      app: nginx-strategy
  template:
    metadata:
      labels:
        app: nginx-strategy
    spec:
      containers:
      - name: nginx
        image: nginx:1.27
        ports:
        - containerPort: 80
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl apply -f nginx-strateg
y-test.yaml
deployment.apps/nginx-strategy-test created
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get deployment nginx-s
trategy-test
NAME                 READY   UP-TO-DATE   AVAILABLE   AGE
nginx-strategy-test   3/3     3            3           11s
ubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get pods -l app=nginx-
strategy
NAME                                  READY   STATUS    RESTARTS   AGE
nginx-strategy-test-588f59556d-8bptz   1/1     Running   0          17s
nginx-strategy-test-588f59556d-clj9d   1/1     Running   0          17s
nginx-strategy-test-588f59556d-lnj8q   1/1     Running   0          17s
```

06 / Bonus Challenge — Strategy Execution & Zero Downtime

Bonus Execution: Observed live pod transitions during image update. All 3 original pods remained fully active until replacement pods were ready, guaranteeing 100% availability.

```
^Cubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl get pods -l app=ngin
x-strategy -w
NAME                                  READY   STATUS    RESTARTS   AGE
nginx-strategy-test-588f59556d-8bptz   1/1     Running   0          47s
nginx-strategy-test-588f59556d-clj9d   1/1     Running   0          47s
nginx-strategy-test-588f59556d-lnj8q   1/1     Running   0          47s
nginx-strategy-test-779cfdc86c-vzc44   0/1     Pending   0          0s
nginx-strategy-test-779cfdc86c-vzc44   0/1     Pending   0          0s
nginx-strategy-test-779cfdc86c-vzc44   0/1     ContainerCreating   0          0s
nginx-strategy-test-779cfdc86c-vzc44   1/1     Running             0          1s
nginx-strategy-test-588f59556d-8bptz   1/1     Terminating         0          116s
nginx-strategy-test-779cfdc86c-p27d9   0/1     Pending             0          0s
nginx-strategy-test-779cfdc86c-p27d9   0/1     Pending             0          0s
nginx-strategy-test-779cfdc86c-p27d9   0/1     ContainerCreating   0          0s
nginx-strategy-test-588f59556d-8bptz   0/1     Terminating         0          117s
nginx-strategy-test-779cfdc86c-p27d9   1/1     Running             0          1s
nginx-strategy-test-588f59556d-8bptz   0/1     Terminating         0          117s
nginx-strategy-test-588f59556d-8bptz   0/1     Terminating         0          117s
nginx-strategy-test-588f59556d-lnj8q   1/1     Terminating         0          117s
nginx-strategy-test-779cfdc86c-pqj67   0/1     Pending             0          0s
nginx-strategy-test-779cfdc86c-pqj67   0/1     Pending             0          0s
nginx-strategy-test-779cfdc86c-pqj67   0/1     ContainerCreating   0          0s
nginx-strategy-test-588f59556d-lnj8q   0/1     Terminating         0          118s
nginx-strategy-test-779cfdc86c-pqj67   1/1     Running             0          1s
nginx-strategy-test-588f59556d-lnj8q   0/1     Terminating         0          118s
nginx-strategy-test-588f59556d-lnj8q   0/1     Terminating         0          118s
nginx-strategy-test-588f59556d-clj9d   1/1     Terminating         0          118s
nginx-strategy-test-588f59556d-clj9d   0/1     Terminating         0          119s
nginx-strategy-test-588f59556d-clj9d   0/1     Terminating         0          119s
nginx-strategy-test-588f59556d-clj9d   0/1     Terminating         0          119s
^Cubuntu@master-node:~/day16-kubernetes-rollout-90-days-DevOps-winter-arc$ kubectl rollout status deplo
deployment "nginx-strategy-test" successfully rolled out
```

```
46 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Mon Oct 19 12:05:35 2026 from 122.183.33.226
ubuntu@master-node:~$ kubectl set image deployment/nginx-strategy-test nginx=nginx:1.28
deployment.apps/nginx-strategy-test image updated
ubuntu@master-node:~$
```

04 / Troubleshooting & Failure Analysis Matrix

Problem / Scenario	Diagnostic Command	Resolution & Key Learning
Rollout status never completes	`kubectl describe pod <pod-name>`	Inspect Pod events for image pull errors, readiness probe failures, or resource bottlenecks.
ImagePullBackOff	`kubectl describe pod` / `kubectl get events`	Confirm correct container image name and tag. Roll back immediately after collecting failure evidence.
History missing release note	`kubectl rollout history`	Ensure `kubernetes.io/change-cause` annotation was explicitly applied before updating the image.
Rollback appears ineffective	`kubectl get pods -o wide`	Verify current image version, wait for rollout status completion, and target specific revisions using `--to-revision`.

07 / Deliverable & Review Criteria Checklist

Repository Structure (`day16-rollouts/`)

- **README.md:** Explains rolling updates, revision history, maxSurge, maxUnavailable, and rollbacks in detail.
- **evidence/rollout-success.png:** Captured successful rollout status and revision history.
- **evidence/failed-rollout-events.png:** Captured failed image update and Pod events (`ImagePullBackOff`).
- **evidence/rollback-verified.png:** Proves rollback restored known-good image and healthy pods.

Security & Review Points (100 Marks)

- **Security:** No kubeconfig files, tokens, or cloud credentials committed.
- **Scoring Breakdown:**
  - Successful update & status (20 pts)
  - Revision history & annotation (20 pts)
  - Failed rollout diagnosed (20 pts)
  - Rollback verified (25 pts) • README/Cleanup (15 pts)

08 / Interview Readiness & Architecture Q&A

1. What does a rolling update do?

- Gradually replaces old Pods with the new Pod template while strictly applying availability limits.

2. What are maxSurge and maxUnavailable?

- Controls extra Pods allowed above desired count and desired Pods allowed to be unavailable during updates.

3. How do you diagnose a failed rollout?

- Inspect Pods, describe failing Pods, and review cluster events or container logs.

4. Why annotate a release with change-cause?

- Records the reason for a change, making deployment revision history easily auditable.